

Local Community Support for Tangible Food Aid During COVID-19

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ABSTRACT

The COVID-19 pandemic is a long-haul disaster that has brought forth extreme hardship, in particular food insecurity. Local citizens across the United States created online groups to facilitate mutual aid, a form of community-based disaster relief. In this paper, we are interested in how online mutual aid groups addressed food insecurity through facilitating various community-based initiatives. We interviewed mutual aid group organizers in the United States and found that online groups contributed immediate food relief as well as laid the groundwork for long-term food security. These findings have implications for community resilience in the wake of an ongoing disaster.

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1 INTRODUCTION & BACKGROUND

The COVID-19 global pandemic was announced in March, 2020 [12] and has brought unprecedented hardship across the world. As of early August, 2021 in the United States, there were over 35 million cases and at least 600,000 deaths attributed to COVID-19 [20]. Unemployment skyrocketed, reaching highs greater than those of the Great Depression [2], and the country grappled with the effects of economic recession [9]. As the pandemic progressed, local government and non-profit resources were stretched increasingly thin, which made it difficult for people experiencing the secondary impacts of COVID-19 (e.g., financial, employment, housing, child-care, healthcare) [15] to access the resources that they need. One of the societal issues the pandemic caused was food insecurity. Due to the high unemployment rate followed COVID-19, people who never experienced food insecurity struggled to purchase food or hot meals. Children who usually received meals from school had to find other means to secure food due to school closures. In response to such social issues, we witnessed a revival of reciprocity as people took it upon themselves to organize and participate in informal networks of neighborly help, called *mutual aid*. Everyday citizens

formed support networks on online platforms and mobilized them in the offline space [6].

Mutual aid is localized, grassroots community-based disaster relief where people exchange information, ask for various kinds of help, and offer to provide help for others, often born from the realization that current top-down systems are not meeting the needs of the community [18]. Similarly, previous work in Computer-supported Cooperative Work (CSCW) explored how local people in affected areas leveraged social media or online communities to exchange tangible support (e.g., food, shelter, rescue activities) or share information (e.g., disaster status, evacuation) during disasters [14, 16, 17, 19]. Although impacts of community-based disaster relief via online platforms have been explored, they are not studied in the context of a global pandemic or a long-haul disaster lasting over a year.

Understanding impacts and strategies of community-based relief during the current pandemic can guide other local communities when another epidemic occurs in the future. This study is the first step of a larger project to understand the impact of online mutual aid as a pillar of local community-based disaster relief during a widespread, long-haul biological disaster. In this study, we are interested in understanding how online mutual aid groups addressed food insecurity, which became prevalent as more and more people were negatively impacted by the secondary affects of the pandemic [10]. To this end, we conducted remote interviews with 17 administrators of online mutual aid groups. Specifically, we seek to answer to following research question: *How are local community members facilitating food aid through online mutual aid during a global pandemic?* Our findings suggest that the online groups contributed food aid for immediate needs and laid the groundwork for long-term food security.

2 METHODS & FINDINGS

We conducted semi-structured interviews with 17 participants, 13 female and four male, who represented 13 COVID-19-related geographic-based (i.e., local area) online mutual aid groups across the United States. All participants were local residents of the area the group covered and were volunteering in an administrator role in their group. 16 of 17 used Facebook as their online platform (Facebook is the most widely-used platform [6]), while one participant created her own website. Each interview lasted approximately one to 1.5 hours and was conducted remotely, either via phone or video conferencing software. Participants from the same group were interviewed together. To discover online mutual aid groups from which to sample, we used the publicly available Mutual Aid

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Hub pandemic-related Mutual Aid Group data set¹ as well as snow-ball sampling. Participants met the following requirements: 1) be an active administrator of an online mutual aid group and 2) consent to a recorded interview. All interviews were audio recorded, transcribed and coded iteratively using the constant comparative analysis method developed by Glaser and Strauss [4]. This study was approved by the university's Institutional Review Board. Our data surfaced two major themes regarding food aid in online mutual aid groups: community-based support for *immediate aid* and initiatives aimed at *long-term aid* for sustainable food security.

Our findings recount how our participant's mutual aid groups provided a safe space for facilitating food aid through a variety of initiatives aimed at supporting both immediate and long-term needs. Almost all of our participants told us that food aid was a frequent request in their group. They made an effort to meet people where they were without judgement for anyone's personal circumstances and to provide culturally-inclusive food support, e.g., making sure that people received familiar food items, which embodies the *self-determination* value of mutual aid [18]. For the first theme, *immediate food aid*, the group provided an online space to facilitate cash donations, grocery delivery, preparation of hot meals, and partnerships with local restaurants. Because COVID-19 is a long-haul disaster, immediate needs became difficult to fulfill over a long period, so our participants came up with ways to use the group to structure food aid for long-term sustainability. These initiatives comprise theme two *long-term food support*, and include community fridges / little free pantries as well as the development of community gardens and food cultivation education. In the following subsections, we provide details into the activities that comprise each theme.

2.1 Immediate Food Aid

Online community-based food aid filled an urgent gap between people who needed food immediately and people who could provide food or food-related support during COVID-19. The food aid organized through the online mutual aid groups differed from that of traditional charities (e.g., food banks), which require people to meet income thresholds in order to access aid. None of our participants means-tested any of the group members because as P9 pointed out, some newly vulnerable people were ashamed to visit a food bank during the pandemic. Our participants told us that many people in their online groups asked for cash to purchase food, particularly if they have food restrictions which make it difficult to accept community food donations. When asked about why cash was popular for food aid requests, P6 explained that if someone "asked for food donation, they're not gonna go give [them] those things [specific food requests]. They're going to give them whatever they can because the places that they're getting food from are just community members donating food. They'd be better off to say 'I really need 25 dollars because I need creamer and some coffee'. So that way they can get the things that they're able to use instead of just something". Most participants shared that their group had a crowd-funded community donation fund, which they used to provide cash to people who requested financial help getting food. For example, P5 said "I've given out about \$20,000 worth of

financial assistance directly to people. And that's either by paying for groceries or supplies that they're requesting [on the news feed], or just sending them money [via a payment app]".

Another way food aid was distributed through the group was through grocery delivery. Especially early on in the pandemic, some people were unable to shop for themselves (i.e., immunocompromised or isolating due to having COVID-19), so they enlisted help from a neighbor by making a post to the group. P8 told us that her group's grocery delivery operation was "started [...] because we quickly realized that was a huge need that was not getting met, particularly folks being able to leave their home to pick up food". P8 and P13 shared Google Forms with group members to arrange food purchases and deliveries for people who didn't have financial resources to buy groceries or were unable to get to a store. P8 also had requesters fill out contact information and a list of food allergies and preferences.

Some participants (i.e., P10, P11, P12 from the same group, as well as P1, P2, and P4 from different groups) mentioned that their groups facilitated food aid through community-organized meal preparation and distribution, all organized through posts on the group. In these cases, the group members responded to the need that they observed on the group's timeline, and donated ingredients, cooked and/or delivered food to the food requesters. For example, P2 told us that each week one member in her group would post about how she "cooks up a couple dozen home cooked meals and goes out up and down Main Street, and hits all of the congregation spots to make sure [houseless] people get at least one hot meal this week". P10, P11, and P12 of the same group prepared food for community meals once a week. They used their group's news feed to share the time and location of the meal so that group members can show up for a hot, home-cooked meal. P1 told us that their group facilitated partnerships with local restaurants to provide urgent hot meals for COVID positive people who were recovering at home. Group members could access these meals by posting to the group's page or contacting an admin through a private message. These initiatives supported immediate hunger relief by providing hot, nutritious food to community members when they needed it.

2.2 Long-term Food Support

The second way that online mutual aid groups facilitated food aid was through long-term relief efforts, which impacted both group members and general members of the local community who were not members of the online group. Across our participants, we learned that this type of aid supported chronic need in their local community through the organization of *little free pantries*² and *community fridges*³ (here-forth referred to as food programs), and empowered residents to cultivate their own fresh food through *container gardens* and *community gardens*.

Participants from seven different groups mentioned that their online group acted as an organizing space for nascent food programs and provided information about pantry or fridge locations in the local area. The online space was vital to facilitating long-term aid because of the interconnected network of helpers that admins could harness to develop food programs (P8), and visibility into trends

¹Available at <https://www.mutualaidhub.org/table-of-networks>

²<https://www.littlefreepantry.org/>

³https://en.wikipedia.org/wiki/Community_fridge

concerning food needs (P2). Due to the overwhelming requests for immediate food aid in her group, P1 initially conceived the idea of a grass-roots local food pantry to provide food for anyone in her community. She said that an autonomous team formed over the group to lead the initiative, and expanded the concept to be culturally inclusive because they “found that the local food banks were not providing things like Maseca, jasmine rice, and things that [their] community needed”. P16 often posted pictures of the little free pantries that her group set up around town that were in need of donations to urge members to “fill them as often as [they] can [...] because people use those things”. Members of P7’s group started a food program that grew to operate over 30 little free pantries in his area. He said “There’s the [color] food pantry cabinet [team] that focuses on coordinating those efforts. The leadership team oversees all the different projects and gets reports and makes some decisions”. They made an auxiliary Facebook Group and a website where local citizens can find a location nearby to donate food or to pick up food if they need it. In P11’s group, food aid requesters could fill out an online form to opt-in to receive a bag of groceries that were purchased with community donations once per week. She mentioned that the program was successful in meeting the goal of weekly grocery bags, and although it would sometimes take time to collect enough donations or price out new items, the group was particularly successful at “supporting special requests for specific grocery items that [recipients] would like to come in the boxes”. While the organization of the food programs and information for how to contribute support or request food aid was facilitated through the online group, the food programs provided an important tangible link to online mutual aid efforts in the local community. Many of the programs were set up in public locations throughout town, making them open to anyone who passed by, not just members of the online group.

Admins in four different groups mentioned organizing local people to cultivate community gardens as a response to food insecurity and the growing number of people who needed food during the pandemic. For P7, community gardens were a form of food activism, whereby growing one’s own food is an act of empowerment as much as it is a source of nourishment. Other admins, such as P4 took a more pragmatic approach: “Last year we did this, and we’ll do again this year, a container gardening project. It was specifically for Black, Indigenous, people of color. Where we built container gardens and passed them out for free [...] we just delivered for people that had transportation issues”.

Education around household food production was also a key part of long-term food aid initiatives. For P16, community gardening was not just about feeding one’s neighbors, but also about growing people’s interest in food cultivation. She said, “we have a community garden plot [...] and that’s something that we’re going to invest a lot of time and effort into. It’s not just getting actual food grown and back out into the community, but connecting people with the process of growing their food, and knowing their food, and getting back into the land”. Most admins told us that they intended to continue advancing their long-term efforts even after the pandemic wanes.

3 DISCUSSION & CONCLUSION

Society faced an acute increase in food insecurity during the pandemic [10]. Aside from food insecurity, some people felt unsafe going grocery shopping due to the risks of contracting the virus. However, restrictions on face-to-face interactions to prevent virus spread made it harder to organize civic disaster relief efforts. Thus, social media was an essential medium for reaching out to people in need and coordinating volunteers [5]. For instance, mutual aid groups shared dates for meal distribution via Facebook. In line with prior research on how social media is a key facilitator of community-based disaster relief (e.g., sharing location-specific information [13], distributing goods and emotional support [3]), during the pandemic we found that online mutual aid was a vital channel for people who can provide various support (e.g., grocery delivery or a hot meal, cooking hot meals) to connect with those who needed support.

Organized by people situated in the same geographic community, these food relief activities were not bound to an online space. In previous disasters, sharing resource information and appreciation through social media strengthened neighborhood social relationships [7]. Building on this, online mutual aid groups inter-weaved online and offline engagement which brought relief to the wider community who may not be aware of the online groups; after organizing online, people physically gathered to realize tangible food aid by distributing hot meals or groceries and filling community pantries. Because of their extension to the offline space, other local community members who might not have Internet or Facebook access could volunteer or benefit from the group’s efforts. Once non-members find out about pantries from their neighbors, they can also stock or grab food from the pantries. Extension of disaster relief to offline space also extends relationships formed online to offline. Possibly due to trust that was developed between people in the group [8], members willingly exchanged contact information to work on initiatives and facilitate aid exchange. Localized offline disaster relief builds weak ties between community members and strengthens community bonding, which is an attribute of resilient communities [1]. These newfound weak ties contribute to rapid, efficient future disaster relief.

Not only did mutual aid groups help people struggling from food insecurity to stay afloat, but they also nourished them with proper humanistic care. First, beyond securing food, mutual aid groups ensured *food dignity* for those in need, emphasizing that people should have food choice. They made sure people could specify preferred foods, get at least one hot meal every week instead of instant or canned food, and enjoy culturally-inclusive food. Second, although COVID-19 caused overwhelming food aid requests due to unexpected life situations (e.g., risks associated with grocery shopping, food insecurity from unemployment), neighbors made sure that fewer people experience hunger. In the group, other members saw that food needs were being met, which suggests neighborly care.

Instead of relying on a centralized food distribution system to come in and distribute food, mutual aid groups developed long-term and self-sufficient community-based means of food security, i.e., community gardens and food cultivation educational programs. Amidst disasters which can cause economic disruption and food shortages, both immediate and sustained access to food is vital.

Communities can address food insecurity independently with their own capacities, and such self-sufficiency will eventually make them more prepared and resilient in future crises [11]. Furthermore, the long-term food aid projects are not confined to disaster relief as admins planned to continue the initiatives after the pandemic stabilizes. Although COVID-19 surfaced the need for hunger relief, people who suffer from food insecurity have always existed. Continuing long-term initiatives can build groundwork for food security and attenuate the root cause of hunger. For example, with community gardens, people have better access to fresh produce.

Our findings provide a foundational *blueprint* for community-based food aid efforts. Other communities can borrow the initiatives discussed in this paper to address specific needs in their own areas, considering their demographics. Additionally, making food aid efforts visible both online and offline can strengthen community-based relief so that the pandemic does not create further suffering and inequality with regard to food security. In the future, we plan to build on this preliminary analysis by examining other types of COVID-19-related needs and relief efforts in social media disaster-relief. Our future studies will analyze posts on online platforms and perform more extensive interviews with group admins to understand further impact of mutual aid on local communities.

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REFERENCES

- [1] Rashid Ahmed, Mohamed Seedat, Ashley Van Niekerk, and Samed Bulbulia. 2004. Discerning community resilience in disadvantaged communities in the context of violence and injury prevention. *South African Journal of Psychology* 34, 3 (2004), 386–408.
- [2] David L. Blustein, Ryan Duffy, Joaquim A. Ferreira, Valerie Cohen-Scali, Rachel Gali Cinamon, and Blake A. Allan. 2020. Unemployment in the time of COVID-19: A research agenda. *Journal of Vocational Behavior* 119 (2020).
- [3] Dharma Dailey, John Robinson, and Kate Starbird. 2016. Sharing food, gathering information: the context and visibility of community information work in a crisis event. *ICConference 2016 Proceedings* (2016).
- [4] Barney G Glaser and Anselm L Strauss. 2017. *Discovery of grounded theory: Strategies for qualitative research*. Routledge.
- [5] Jeongwon Jo, Tiffany Kneareem, Chun-Hua Tsai, and John M Carroll. 2021. COVID-19 Kindness: Patterns of Neighborly Cooperation during a Global Pandemic. In *C&T'21: Proceedings of the 10th International Conference on Communities & Technologies-Wicked Problems in the Age of Tech*. 1–14.
- [6] Tiffany Kneareem, Jeongwon Jo, Chun-Hua Tsai, and John M Carroll. 2021. Making Space for Support: An Exploratory Analysis of Pandemic-Response Mutual Aid Platforms. In *C&T'21: Proceedings of the 10th International Conference on Communities & Technologies-Wicked Problems in the Age of Tech*. 38–43.
- [7] Jing Li, Keri K Stephens, Yaguang Zhu, and Dhiraj Murthy. 2019. Using social media to call for help in Hurricane Harvey: Bonding emotion, culture, and community relationships. *International Journal of Disaster Risk Reduction* 38 (2019), 101212.
- [8] Wendy Macias, Karen Hilyard, and Vicki Freimuth. 2009. Blog functions as risk and crisis communication during Hurricane Katrina. *Journal of computer-mediated communication* 15, 1 (2009), 1–31.
- [9] Maria Nicola, Zaid Alsafi, Catrin Sohrabi, Ahmed Kerwan, Ahmed Al-Jabir, Christos Iosifidis, Maliha Agha, and Riaz Agha. 2020. The socio-economic implications of the coronavirus pandemic (COVID-19): A review. *International journal of surgery (London, England)* 78 (2020), 185.
- [10] Meredith T Niles, Farryl Bertmann, Emily H Belarmino, Thomas Wentworth, Erin Biehl, and Roni Neff. 2020. The early food insecurity impacts of COVID-19. *Nutrients* 12, 7 (2020), 2096.
- [11] Fran H Norris, Susan P Stevens, Betty Pfefferbaum, Karen F Wyche, and Rose L Pfefferbaum. 2008. Community resilience as a metaphor, theory, set of capacities, and strategy for disaster readiness. *American journal of community psychology* 41, 1 (2008), 127–150.
- [12] World Health Organization. 2020. *Listings of WHO's response to COVID-19*. <https://www.who.int/news/item/29-06-2020-covidtimeline>
- [13] Leysia Palen, Starr Roxanne Hiltz, and Sophia B Liu. 2007. Online forums supporting grassroots participation in emergency preparedness and response. *Commun. ACM* 50, 3 (2007), 54–58.
- [14] Yan Qu, Chen Huang, Pengyi Zhang, and Jun Zhang. 2011. Microblogging after a major disaster in China: a case study of the 2010 Yushu earthquake. In *Proceedings of the ACM 2011 conference on Computer supported cooperative work*. 25–34.
- [15] Alexandra Quinn and Margaret Laws. 2020. Addressing community needs and preparing for the secondary impacts of Covid-19. *Nejm Catalyst Innovations in Care Delivery* (2020).
- [16] Bryan Semaan and Gloria Mark. 2012. 'Facebooking' towards crisis recovery and beyond: Disruption as an opportunity. In *Proceedings of the ACM 2012 conference on computer supported cooperative work*. 27–36.
- [17] Irina Shklovski, Leysia Palen, and Jeannette Sutton. 2008. Finding community through information and communication technology in disaster response. In *Proceedings of the 2008 ACM conference on Computer supported cooperative work*. 127–136.
- [18] Dean Spade. 2020. *Mutual Aid: Building Solidarity during this Crisis (and the Next)*. Verso Books.
- [19] Kate Starbird, Leysia Palen, Amanda L Hughes, and Sarah Vieweg. 2010. Chatter on the red: what hazards threat reveals about the social life of microblogged information. In *Proceedings of the 2010 ACM conference on Computer supported cooperative work*. 241–250.
- [20] Worldometer. 2021. *United States Coronavirus Cases*. <https://www.worldometers.info/coronavirus/country/us/>